

```

import pandas as pd
from google.colab import files
import seaborn as sns
import matplotlib.pyplot as plt

# Upload the file to the current session
uploaded = files.upload()

# Get the filename from the uploaded dictionary
filename = list(uploaded.keys())[0] # Assuming only one file is
uploaded

# Read the CSV file using the uploaded filename
dataset = pd.read_csv(filename)

```

<IPython.core.display.HTML object>

Saving archive (2).zip to archive (2) (1).zip

```
dataset.head(5)
```

```

{"summary":{"\n  \"name\": \"dataset\",\n  \"rows\": 527,\n  \"fields\": [\n    {\n      \"column\": \"order_id\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 73,\n        \"min\": 1,\n        \"max\": 273,\n        \"num_unique_values\": 273,\n        \"samples\": [\n          31,\n          118,\n          80\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"order_date\",\n      \"properties\": {\n        \"dtype\": \"object\",\n        \"num_unique_values\": 444,\n        \"samples\": [\n          \"2022/9/15 4:55:00\",\n          \"2022/7/27 21:35:00\",\n          \"2022/6/9 20:16:00\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"sku\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 24,\n        \"samples\": [\n          \"127\",\n          \"1499\",\n          \"708\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"color\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 39,\n        \"samples\": [\n          \"Mocha\",\n          \"Yellow\",\n          \"Blue\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"size\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 9,\n        \"samples\": [\n          \"S\",\n          \"One Size\",\n          \"L\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"unit_price\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 34,\n        \"min\": 191,\n        \"max\": 899,\n        \"num_unique_values\": 35,\n

```

```

\"samples\": [\n          198,\n          227,\n          318\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      {\n        \"column\": \"quantity\",\n        \"properties\": {\n          \"dtype\": \"number\",\n          \"std\": 0,\n          \"min\": 1,\n          \"max\": 2,\n          \"num_unique_values\": 2,\n          \"samples\": [\n            2,\n            1\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n        },\n        {\n          \"column\": \"revenue\",\n          \"properties\": {\n            \"dtype\": \"number\",\n            \"std\": 44,\n            \"min\": 191,\n            \"max\": 899,\n            \"num_unique_values\": 38,\n            \"samples\": [\n              530,\n              286\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\"\n          }\n        }\n      ]\n    },\n    \"type\": \"dataframe\",\n    \"variable_name\": \"dataset\"}

```

```
dataset.isnull()
```

```

{\"summary\": {\n  \"name\": \"dataset\",\n  \"rows\": 527,\n  \"fields\": [\n    {\n      \"column\": \"order_id\",\n      \"properties\": {\n        \"dtype\": \"boolean\",\n        \"num_unique_values\": 1,\n        \"samples\": [\n          false\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      {\n        \"column\": \"order_date\",\n        \"properties\": {\n          \"dtype\": \"boolean\",\n          \"num_unique_values\": 1,\n          \"samples\": [\n            false\n          ],\n          \"semantic_type\": \"\",\n          \"description\": \"\"\n        },\n        {\n          \"column\": \"sku\",\n          \"properties\": {\n            \"dtype\": \"boolean\",\n            \"num_unique_values\": 1,\n            \"samples\": [\n              false\n            ],\n            \"semantic_type\": \"\",\n            \"description\": \"\"\n          },\n          {\n            \"column\": \"color\",\n            \"properties\": {\n              \"dtype\": \"boolean\",\n              \"num_unique_values\": 1,\n              \"samples\": [\n                false\n              ],\n              \"semantic_type\": \"\",\n              \"description\": \"\"\n            },\n            {\n              \"column\": \"size\",\n              \"properties\": {\n                \"dtype\": \"boolean\",\n                \"num_unique_values\": 2,\n                \"samples\": [\n                  true\n                ],\n                \"semantic_type\": \"\",\n                \"description\": \"\"\n              },\n              {\n                \"column\": \"unit_price\",\n                \"properties\": {\n                  \"dtype\": \"boolean\",\n                  \"num_unique_values\": 1,\n                  \"samples\": [\n                    false\n                  ],\n                  \"semantic_type\": \"\",\n                  \"description\": \"\"\n                },\n                {\n                  \"column\": \"quantity\",\n                  \"properties\": {\n                    \"dtype\": \"boolean\",\n                    \"num_unique_values\": 1,\n                    \"samples\": [\n                      false\n                    ],\n                    \"semantic_type\": \"\",\n                    \"description\": \"\"\n                  },\n                  {\n                    \"column\": \"revenue\",\n                    \"properties\": {\n                      \"dtype\": \"boolean\",\n                      \"num_unique_values\": 1,\n                      \"samples\": [\n                        false\n                      ],\n                      \"semantic_type\": \"\",\n                      \"description\": \"\"\n                    }\n                  }\n                }\n              }\n            }\n          ]\n        }\n      }\n    }\n  ]\n}

```

```
\n\"semantic_type\": \"\", \n      \"description\": \"\" \n    }\n  ]\n} \", \"type\": \"dataframe\"}
```

```
dataset.describe()
```

```
{\"summary\": \"{ \n  \"name\": \"dataset\", \n  \"rows\": 8, \n  \"fields\": [\n    {\n      \"column\": \"order_id\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 167.27399350672368, \n        \"min\": 1.0, \n        \"max\": 527.0, \n        \"num_unique_values\": 8, \n        \"samples\": [\n          110.0, \n          527.0 \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      }, \n      {\n        \"column\": \"unit_price\", \n        \"properties\": {\n          \"dtype\": \"number\", \n          \"std\": 261.59889480166225, \n          \"min\": 34.517412481489515, \n          \"max\": 899.0, \n          \"num_unique_values\": 8, \n          \"samples\": [\n            274.9734345351044, \n            278.0, \n            527.0 \n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\" \n        }, \n        {\n          \"column\": \"quantity\", \n          \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": 185.9638350961581, \n            \"min\": 0.10619310200573744, \n            \"max\": 527.0, \n            \"num_unique_values\": 5, \n            \"samples\": [\n              1.0113851992409868, \n              2.0, \n              0.10619310200573744 \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n          }, \n          {\n            \"column\": \"revenue\", \n            \"properties\": {\n              \"dtype\": \"number\", \n              \"std\": 259.7687986335217, \n              \"min\": 44.74109464499173, \n              \"max\": 899.0, \n              \"num_unique_values\": 8, \n              \"samples\": [\n                278.0246679316888, \n                278.0, \n                527.0 \n              ], \n              \"semantic_type\": \"\", \n              \"description\": \"\" \n            } \n          } \n        ] \n      }, \n      {\n        \"column\": \"quantity\", \n        \"properties\": {\n          \"dtype\": \"number\", \n          \"std\": 185.9638350961581, \n          \"min\": 0.10619310200573744, \n          \"max\": 527.0, \n          \"num_unique_values\": 5, \n          \"samples\": [\n            1.0113851992409868, \n            2.0, \n            0.10619310200573744 \n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\" \n        }, \n        {\n          \"column\": \"revenue\", \n          \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": 259.7687986335217, \n            \"min\": 44.74109464499173, \n            \"max\": 899.0, \n            \"num_unique_values\": 8, \n            \"samples\": [\n              278.0246679316888, \n              278.0, \n              527.0 \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n          } \n        } \n      } \n    ] \n  } \", \"type\": \"dataframe\"}
```

```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 527 entries, 0 to 526
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  -
0   order_id    527 non-null    int64
1   order_date  527 non-null    object
2   sku         527 non-null    object
3   color       527 non-null    object
4   size        490 non-null    object
5   unit_price  527 non-null    int64
6   quantity    527 non-null    int64
7   revenue     527 non-null    int64
dtypes: int64(4), object(4)
memory usage: 33.1+ KB
```

```
dataset.isnull().sum()
```

```
order_id      0
order_date    0
sku           0
color         0
size         37
unit_price    0
quantity      0
revenue       0
dtype: int64
```

```
import pandas as pd
import numpy as np
from google.colab import files
import seaborn as sns
import matplotlib.pyplot as plt
```

```
# Upload the file to the current session
```

```
uploaded = files.upload()
```

```
# Get the filename from the uploaded dictionary
```

```
filename = list(uploaded.keys())[0] # Assuming only one file is
uploaded
```

```
# Read the CSV file using the uploaded filename
```

```
dataset = pd.read_csv(filename)
```

```
<IPython.core.display.HTML object>
```

```
Saving archive (2).zip to archive (2) (2).zip
```

```
dataset.head(5)
```

```
{"summary":"{\n  \"name\": \"dataset\", \n  \"rows\": 527, \n  \"fields\": [\n    {\n      \"column\": \"order_id\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 73, \n        \"min\": 1, \n        \"max\": 273, \n        \"num_unique_values\": 273, \n        \"samples\": [\n          31, \n          118, \n          80\n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    }, \n    {\n      \"column\": \"order_date\", \n      \"properties\": {\n        \"dtype\": \"object\", \n        \"num_unique_values\": 444, \n        \"samples\": [\n          \"2022/9/15 4:55:00\", \n          \"2022/7/27 21:35:00\", \n          \"2022/6/9 20:16:00\" \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    }, \n    {\n      \"column\": \"sku\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 24, \n        \"samples\": [\n          \"127\", \n          \"1499\", \n          \"708\" \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    }, \n    {\n      \"column\": \"color\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 10, \n        \"samples\": [\n          \"black\", \n          \"white\", \n          \"red\", \n          \"blue\", \n          \"green\", \n          \"yellow\", \n          \"purple\", \n          \"brown\", \n          \"pink\", \n          \"grey\" \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    }, \n    {\n      \"column\": \"size\", \n      \"properties\": {\n        \"dtype\": \"category\", \n        \"num_unique_values\": 10, \n        \"samples\": [\n          \"XS\", \n          \"S\", \n          \"M\", \n          \"L\", \n          \"XL\", \n          \"XXL\", \n          \"XXXL\", \n          \"4XL\", \n          \"5XL\", \n          \"6XL\" \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    }, \n    {\n      \"column\": \"unit_price\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 1.5, \n        \"min\": 0.5, \n        \"max\": 3.5, \n        \"num_unique_values\": 10, \n        \"samples\": [\n          0.5, \n          1.0, \n          1.5, \n          2.0, \n          2.5, \n          3.0, \n          3.5, \n          4.0, \n          4.5, \n          5.0 \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    }, \n    {\n      \"column\": \"quantity\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 1.0, \n        \"min\": 1, \n        \"max\": 10, \n        \"num_unique_values\": 10, \n        \"samples\": [\n          1, \n          2, \n          3, \n          4, \n          5, \n          6, \n          7, \n          8, \n          9, \n          10 \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    }, \n    {\n      \"column\": \"revenue\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 1.5, \n        \"min\": 0.5, \n        \"max\": 3.5, \n        \"num_unique_values\": 10, \n        \"samples\": [\n          0.5, \n          1.0, \n          1.5, \n          2.0, \n          2.5, \n          3.0, \n          3.5, \n          4.0, \n          4.5, \n          5.0 \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      } \n    } \n  ], \n  \"description\": \"\" \n}
```

```

{"color": "\n      \"properties\": {\n          \"dtype\": \"category\", \n          \"num_unique_values\": 39, \n          \"samples\": [\n            \"Mocha\", \n            \"Yellow\", \n            \"Blue\" \n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\" \n        }, \n        {\n          \"column\": \"size\", \n          \"properties\": {\n            \"dtype\": \"category\", \n            \"num_unique_values\": 9, \n            \"samples\": [\n              \"S\", \n              \"One Size\" \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n          }, \n          {\n            \"column\": \"unit_price\", \n            \"properties\": {\n              \"dtype\": \"number\", \n              \"std\": 34, \n              \"min\": 191, \n              \"max\": 899, \n              \"num_unique_values\": 35, \n              \"samples\": [\n                198, \n                227, \n                318 \n              ], \n              \"semantic_type\": \"\", \n              \"description\": \"\" \n            }, \n            {\n              \"column\": \"quantity\", \n              \"properties\": {\n                \"dtype\": \"number\", \n                \"std\": 0, \n                \"min\": 1, \n                \"max\": 2, \n                \"num_unique_values\": 2, \n                \"samples\": [\n                  1 \n                ], \n                \"semantic_type\": \"\", \n                \"description\": \"\" \n              }, \n              {\n                \"column\": \"revenue\", \n                \"properties\": {\n                  \"dtype\": \"number\", \n                  \"std\": 44, \n                  \"min\": 191, \n                  \"max\": 899, \n                  \"num_unique_values\": 38, \n                  \"samples\": [\n                    530, \n                    286 \n                  ], \n                  \"semantic_type\": \"\", \n                  \"description\": \"\" \n                } \n              } \n            } \n          ], \n          \"type\": \"dataframe\", \n          \"variable_name\": \"dataset\"}

```

dataset.describe()

```

{"summary": "{\n  \"name\": \"dataset\", \n  \"rows\": 8, \n  \"fields\": [\n    {\n      \"column\": \"order_id\", \n      \"properties\": {\n        \"dtype\": \"number\", \n        \"std\": 167.27399350672368, \n        \"min\": 1.0, \n        \"max\": 527.0, \n        \"num_unique_values\": 8, \n        \"samples\": [\n          115.31309297912713, \n          110.0, \n          527.0 \n        ], \n        \"semantic_type\": \"\", \n        \"description\": \"\" \n      }, \n      {\n        \"column\": \"unit_price\", \n        \"properties\": {\n          \"dtype\": \"number\", \n          \"std\": 261.59889480166225, \n          \"min\": 34.517412481489515, \n          \"max\": 899.0, \n          \"num_unique_values\": 8, \n          \"samples\": [\n            274.9734345351044, \n            278.0, \n            527.0 \n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\" \n        }, \n        {\n          \"column\": \"quantity\", \n          \"properties\": {\n            \"dtype\": \"number\", \n            \"std\": 185.9638350961581, \n            \"min\": 0.10619310200573744, \n            \"max\": 527.0, \n            \"num_unique_values\": 5, \n            \"samples\": [\n              1.0113851992409868, \n              2.0, \n              0.10619310200573744 \n            ], \n            \"semantic_type\": \"\", \n            \"description\": \"\" \n          }, \n          {\n            \"column\": \"revenue\", \n            \"properties\": {\n              \"dtype\": \"number\", \n              \"std\": 44, \n              \"min\": 191, \n              \"max\": 899, \n              \"num_unique_values\": 38, \n              \"samples\": [\n                530, \n                286 \n              ], \n              \"semantic_type\": \"\", \n              \"description\": \"\" \n            } \n          } \n        ] \n      }, \n      {\n        \"column\": \"revenue\", \n        \"properties\": {\n          \"dtype\": \"number\", \n          \"std\": 44, \n          \"min\": 191, \n          \"max\": 899, \n          \"num_unique_values\": 38, \n          \"samples\": [\n            530, \n            286 \n          ], \n          \"semantic_type\": \"\", \n          \"description\": \"\" \n        } \n      } \n    ] \n  }, \n  \"type\": \"dataframe\", \n  \"variable_name\": \"dataset\"}

```

```

n      \ "std\ ": 259.7687986335217,\n      \ "min\ ":
44.74109464499173,\n      \ "max\ ": 899.0,\n
\ "num_unique_values\ ": 8,\n      \ "samples\ ": [\n
278.0246679316888,\n      278.0,\n      527.0\n      ],\n
\ "semantic_type\ ": \ "\",\n      \ "description\ ": \ "\n      }\n
n      }\n      ]\n      }","type":"dataframe"}

```

```
dataset.isnull()
```

```

{"summary":{"\n  \ "name\ ": \ "dataset\ ",\n  \ "rows\ ": 527,\n  \ "fields\ ": [\n    {\n      \ "column\ ": \ "order_id\ ",\n      \ "properties\ ": {\n        \ "dtype\ ": \ "boolean\ ",\n        \ "num_unique_values\ ": 1,\n        \ "samples\ ": [\n          false\n        ],\n        \ "semantic_type\ ": \ "\",\n        \ "description\ ": \ "\n      }\n    },\n    {\n      \ "column\ ": \ "order_date\ ",\n      \ "properties\ ": {\n        \ "dtype\ ": \ "boolean\ ",\n        \ "num_unique_values\ ": 1,\n        \ "samples\ ": [\n          false\n        ],\n        \ "semantic_type\ ": \ "\",\n        \ "description\ ": \ "\n      }\n    },\n    {\n      \ "column\ ": \ "sku\ ",\n      \ "properties\ ": {\n        \ "dtype\ ": \ "boolean\ ",\n        \ "num_unique_values\ ": 1,\n        \ "samples\ ": [\n          false\n        ],\n        \ "semantic_type\ ": \ "\",\n        \ "description\ ": \ "\n      }\n    },\n    {\n      \ "column\ ": \ "color\ ",\n      \ "properties\ ": {\n        \ "dtype\ ": \ "boolean\ ",\n        \ "num_unique_values\ ": 1,\n        \ "samples\ ": [\n          false\n        ],\n        \ "semantic_type\ ": \ "\",\n        \ "description\ ": \ "\n      }\n    },\n    {\n      \ "column\ ": \ "size\ ",\n      \ "properties\ ": {\n        \ "dtype\ ": \ "boolean\ ",\n        \ "num_unique_values\ ": 2,\n        \ "samples\ ": [\n          true\n        ],\n        \ "semantic_type\ ": \ "\",\n        \ "description\ ": \ "\n      }\n    },\n    {\n      \ "column\ ": \ "unit_price\ ",\n      \ "properties\ ": {\n        \ "dtype\ ": \ "boolean\ ",\n        \ "num_unique_values\ ": 1,\n        \ "samples\ ": [\n          false\n        ],\n        \ "semantic_type\ ": \ "\",\n        \ "description\ ": \ "\n      }\n    },\n    {\n      \ "column\ ": \ "quantity\ ",\n      \ "properties\ ": {\n        \ "dtype\ ": \ "boolean\ ",\n        \ "num_unique_values\ ": 1,\n        \ "samples\ ": [\n          false\n        ],\n        \ "semantic_type\ ": \ "\",\n        \ "description\ ": \ "\n      }\n    },\n    {\n      \ "column\ ": \ "revenue\ ",\n      \ "properties\ ": {\n        \ "dtype\ ": \ "boolean\ ",\n        \ "num_unique_values\ ": 1,\n        \ "samples\ ": [\n          false\n        ],\n        \ "semantic_type\ ": \ "\",\n        \ "description\ ": \ "\n      }\n    }\n  ]\n  }","type":"dataframe"}

```

```
dataset.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 527 entries, 0 to 526
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype

```

```

---
0  order_id      527 non-null  int64
1  order_date    527 non-null  object
2  sku           527 non-null  object
3  color         527 non-null  object
4  size          490 non-null  object
5  unit_price    527 non-null  int64
6  quantity      527 non-null  int64
7  revenue       527 non-null  int64

```

dtypes: int64(4), object(4)

memory usage: 33.1+ KB

```
dataset.isnull().sum()
```

```

order_id      0
order_date    0
sku           0
color         0
size          37
unit_price     0
quantity      0
revenue       0
dtype: int64

```

```
dataset.isnull().sum().sum()
```

```
np.int64(37)
```

```

missing_rows = dataset[dataset.isnull().any(axis=1)]
print(missing_rows)

```

	order_id	order_date	sku	color	size	unit_price
quantity \						
9	9	2022/6/11 22:53:00	239	Black	NaN	258
1						
20	18	2022/6/15 13:36:00	77	B	NaN	258
1						
21	18	2022/6/15 13:36:00	77	C	NaN	258
1						
22	18	2022/6/15 13:36:00	77	D	NaN	258
1						
23	18	2022/6/15 13:36:00	77	E	NaN	258
1						
24	18	2022/6/15 13:36:00	77	A	NaN	288
1						
66	55	2022/6/30 12:26:00	439	Rust	NaN	258
1						
68	57	2022/6/5 7:18:00	77	B	NaN	258
1						
81	69	2022/6/6 3:56:00	77	B	NaN	258
1						

82	69	2022/6/6 3:56:00	77	D	NaN	258
1						
83	69	2022/6/6 3:56:00	77	A	NaN	288
1						
137	115	2022/7/21 17:56:00	77	C	NaN	239
1						
193	167	2022/7/7 11:56:00	77	B	NaN	239
1						
206	178	2022/7/9 21:20:00	77	A	NaN	239
1						
233	204	2022/8/15 19:15:00	77	C	NaN	239
1						
234	204	2022/8/15 19:15:00	239	Green	NaN	245
1						
235	204	2022/8/15 19:15:00	239	White	NaN	245
1						
236	204	2022/8/15 19:15:00	239	Mustard	NaN	245
1						
237	204	2022/8/15 19:15:00	239	Rust	NaN	245
1						
238	204	2022/8/15 19:15:00	339	Brown	NaN	254
1						
239	204	2022/8/15 19:15:00	339	Green	NaN	254
1						
282	244	2022/8/25 22:07:00	239	Gray	NaN	278
1						
325	11	2022/8/4 12:30:00	77	B	NaN	239
1						
326	11	2022/8/4 12:30:00	77	C	NaN	239
1						
335	17	2022/8/6 11:04:00	339	Navy Blue	NaN	288
1						
336	17	2022/8/6 11:04:00	339	Mocha	NaN	288
1						
346	25	2022/8/8 15:11:00	77	B	NaN	239
1						
347	25	2022/8/8 15:11:00	239	Navy Blue	NaN	239
1						
348	25	2022/8/8 15:11:00	239	Brown	NaN	239
1						
349	25	2022/8/8 15:11:00	239	Skintone	NaN	278
1						
369	45	2022/9/10 18:43:00	239	Navy Blue	NaN	278
1						
407	75	2022/9/17 21:03:00	239	Rust	NaN	278
1						
414	81	2022/9/18 8:12:00	239	Yellow	NaN	278
1						
443	106	2022/9/24 15:57:00	239	Gray	NaN	259

1							
458	121	2022/9/29 21:14:00	239	Gray	NaN	278	
1							
463	125	2022/9/3 22:13:00	239	Navy Blue	NaN	278	
1							
489	138	2022/9/4 22:57:00	239	Black	NaN	278	
1							

revenue

9	258
20	258
21	258
22	258
23	258
24	288
66	258
68	258
81	258
82	258
83	288
137	239
193	239
206	239
233	239
234	245
235	245
236	245
237	245
238	254
239	254
282	278
325	239
326	239
335	288
336	288
346	239
347	239
348	239
349	278
369	278
407	278
414	278
443	259
458	278
463	278
489	278

convert order_date to datetiem format

```
dataset['order_date'] = pd.to_datetime(dataset['order_date'])
```

check basic info about the dataset

```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 527 entries, 0 to 526
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  -
 0   order_id    527 non-null    int64
 1   order_date  527 non-null    datetime64[ns]
 2   sku         527 non-null    object
 3   color       527 non-null    object
 4   size       490 non-null    object
 5   unit_price  527 non-null    int64
 6   quantity   527 non-null    int64
 7   revenue     527 non-null    int64
dtypes: datetime64[ns](1), int64(4), object(3)
memory usage: 33.1+ KB
```

```
dataset.head(5)
```

```
{"summary":{"\n  \"name\": \"dataset\",\n  \"rows\": 527,\n  \"fields\": [\n    {\n      \"column\": \"order_id\",\n      \"properties\": {\n        \"dtype\": \"number\",\n        \"std\": 73,\n        \"min\": 1,\n        \"max\": 273,\n        \"num_unique_values\": 273,\n        \"samples\": [\n          31,\n          118,\n          80\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"order_date\",\n      \"properties\": {\n        \"dtype\": \"date\",\n        \"min\": \"2022-06-01 16:05:00\",\n        \"max\": \"2022-09-30 23:25:00\",\n        \"num_unique_values\": 444,\n        \"samples\": [\n          \"2022-09-15 04:55:00\",\n          \"2022-07-27 21:35:00\",\n          \"2022-06-09 20:16:00\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"sku\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 24,\n        \"samples\": [\n          \"127\",\n          \"1499\",\n          \"708\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"color\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 39,\n        \"samples\": [\n          \"Mocha\",\n          \"Yellow\",\n          \"Blue\"\n        ],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n      },\n      \"column\": \"size\",\n      \"properties\": {\n        \"dtype\": \"category\",\n        \"num_unique_values\": 9,\n        \"samples\": [\n          \"S\",\n          \"One Size\",\n          \"L\"\n        ],\n        \"semantic_type\": \"\",
```

```

{"description": "\n\n    }\n  },\n  {\n    \"column\":\n    \"unit_price\",\n    \"properties\": {\n      \"dtype\":\n      \"number\",\n      \"std\": 34,\n      \"min\": 191,\n      \"max\": 899,\n      \"num_unique_values\": 35,\n      \"samples\": [\n        198,\n        227,\n        318\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\n\n    }\n  },\n  {\n    \"column\": \"quantity\",\n    \"properties\": {\n      \"dtype\": \"number\",\n      \"std\":\n      0,\n      \"min\": 1,\n      \"max\": 2,\n      \"num_unique_values\": 2,\n      \"samples\": [\n        2,\n        1\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\n\n    }\n  },\n  {\n    \"column\":\n    \"revenue\",\n    \"properties\": {\n      \"dtype\": \"number\",\n      \"std\": 44,\n      \"min\": 191,\n      \"max\": 899,\n      \"num_unique_values\": 38,\n      \"samples\": [\n        530,\n        286\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\n\n    }\n  }\n]\n}","type":"dataframe","variable_name":"dataset"}

```

Total revenue in teh dataset

```

Total_revenue = dataset['revenue'].sum()
print(Total_revenue)

146519

```

Revenue by Month

```

monthly_revenue = dataset.resample('M', on='order_date')
['revenue'].sum()
print(monthly_revenue)

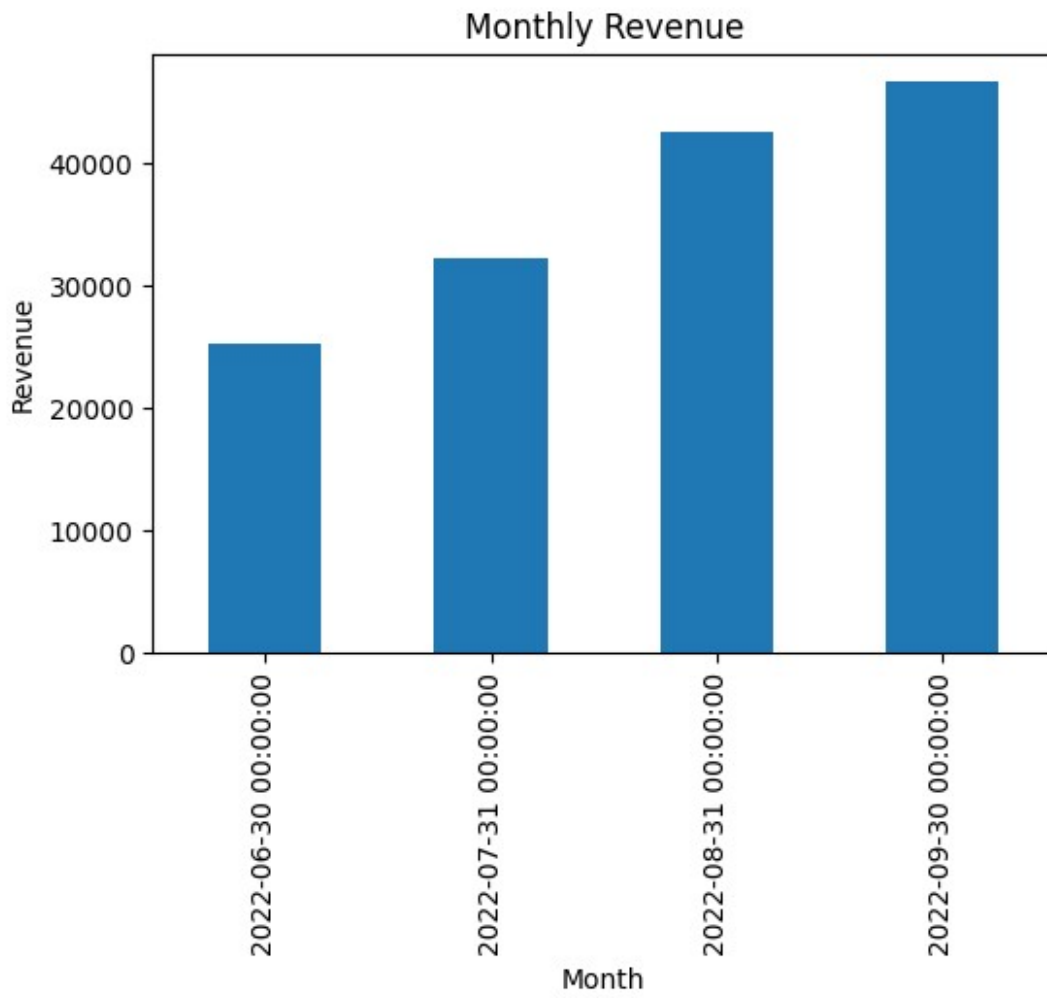
order_date
2022-06-30    25259
2022-07-31    32175
2022-08-31    42513
2022-09-30    46572
Freq: ME, Name: revenue, dtype: int64

<ipython-input-23-8534aa70854a>:1: FutureWarning: 'M' is deprecated
and will be removed in a future version, please use 'ME' instead.
  monthly_revenue = dataset.resample('M', on='order_date')
['revenue'].sum()

monthly_revenue.plot(kind='bar', figsize=(6, 4))
plt.xlabel('Month')
plt.ylabel('Revenue')

```

```
plt.title('Monthly Revenue')
plt.show()
```



Top 10 best- selling products by quantity

```
top_products = dataset.groupby('sku')['quantity'].sum().nlargest(10)
print(top_products)
```

sku	
799	287
708	99
9699	17
bobo	17
239	16
77	16
89	13
3799	9
79	9

```
1719      8
Name: quantity, dtype: int64
```

Revenue by product size

```
size_product = dataset.groupby('size')['revenue'].sum()
print(size_product)

size
2XL      12422
3XL      11301
4XL       3205
L        25362
M        25755
One Size    990
One Size   11793
S         227
XL       45868
Name: revenue, dtype: int64
```

Analyze Revenue by product color

```
color_revernue= dataset.groupby('color')['revenue'].sum()
print(color_revernue)

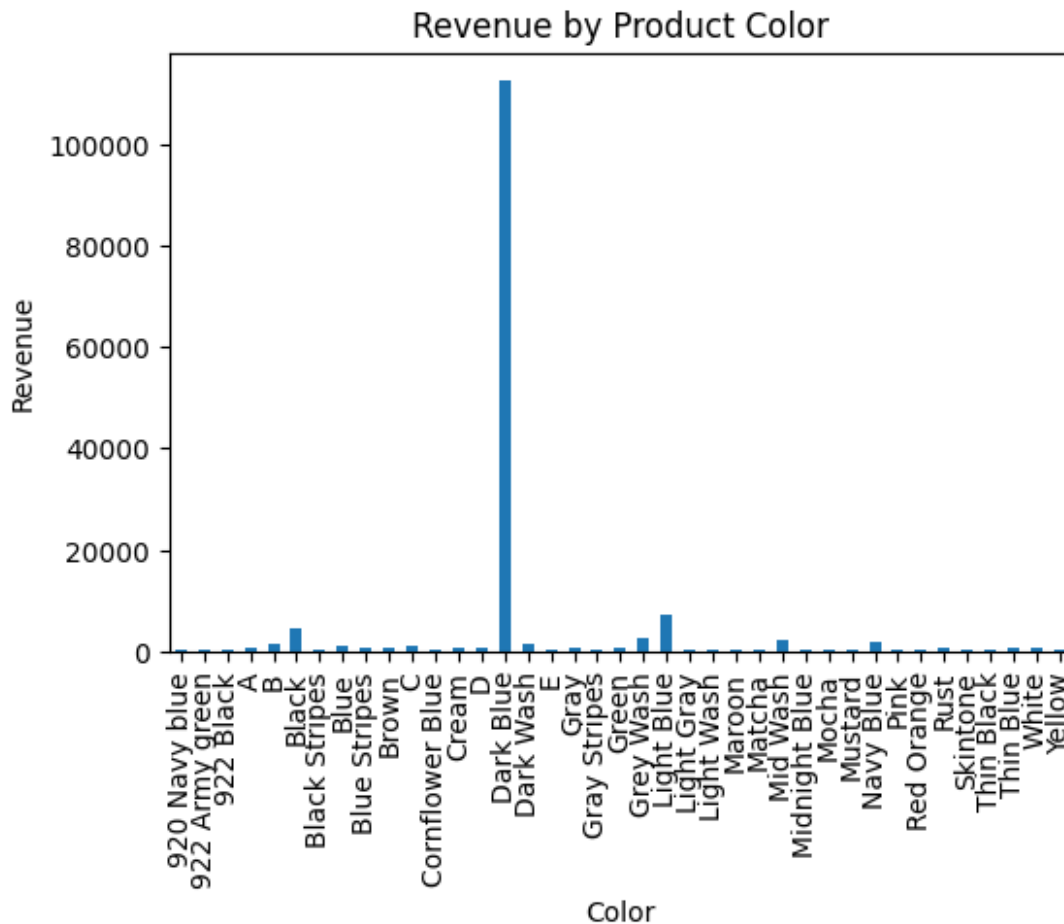
color
920 Navy blue      227
922 Army green     207
922 Black          207
A                  815
B                 1491
Black             4428
Black Stripes     248
Blue              878
Blue Stripes      496
Brown             691
C                 975
Cornflower Blue   198
Cream             610
D                 516
Dark Blue        112364
Dark Wash        1311
E                258
Gray             815
Gray Stripes     255
Green            499
Grey Wash        2390
Light Blue       7337
Light Gray       228
```

Light Wash	238
Maroon	191
Matcha	456
Mid Wash	2043
Midnight Blue	396
Mocha	288
Mustard	245
Navy Blue	1693
Pink	228
Red Orange	228
Rust	781
Skintone	278
Thin Black	268
Thin Blue	785
White	679
Yellow	278

Name: revenue, dtype: int64

visulize it using bar chart

```
color_revernue.plot(kind='bar', figsize=(6, 4))
plt.xlabel('Color')
plt.ylabel('Revenue')
plt.title('Revenue by Product Color')
plt.show()
```



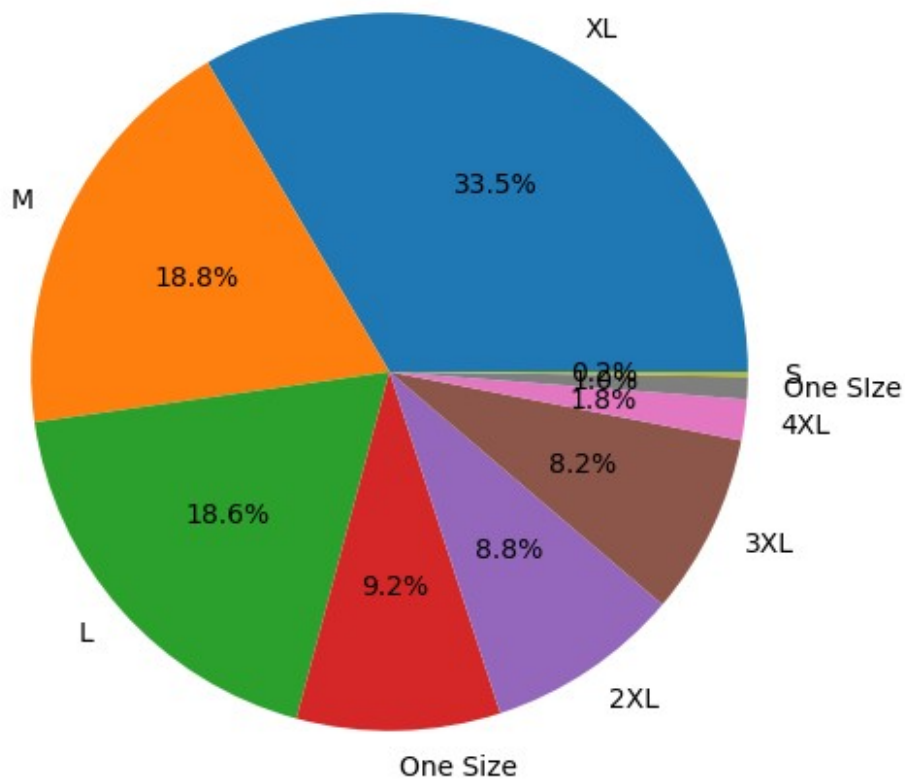
Find the most common sizes sold

```
size_quantity = dataset['size'].value_counts()
print(size_quantity)
```

```
size
XL      164
M       92
L       91
One Size  45
2XL     43
3XL     40
4XL      9
One Size  5
S        1
Name: count, dtype: int64
```

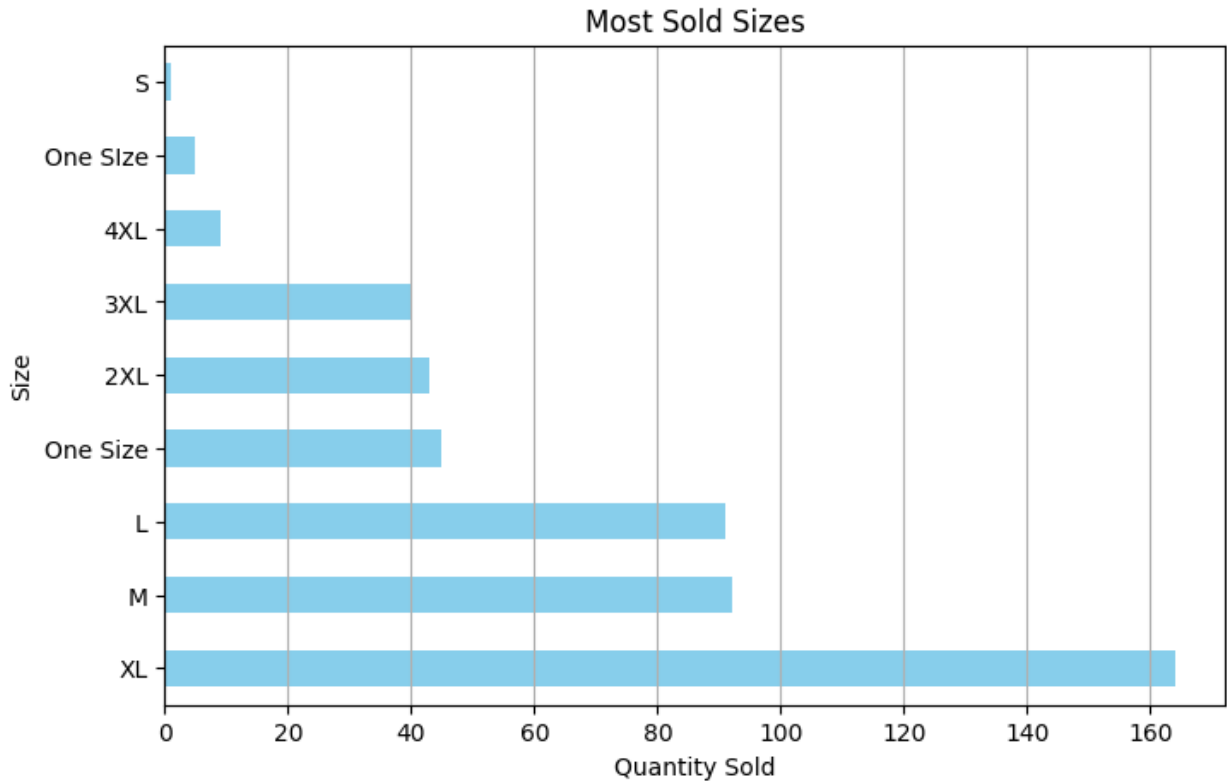
```
size_quantity.plot(kind='pie', autopct='%1.1f%%', figsize=(6, 6),
title="Size Distribution")
plt.ylabel('') # Removes the y-axis label
plt.show()
```

Size Distribution



visulize with a pie chart

```
size_quantity.plot(kind='barh', figsize=(8, 5), color='skyblue',  
title="Most Sold Sizes")  
plt.xlabel('Quantity Sold')  
plt.ylabel('Size')  
plt.grid(axis='x')  
plt.show()
```

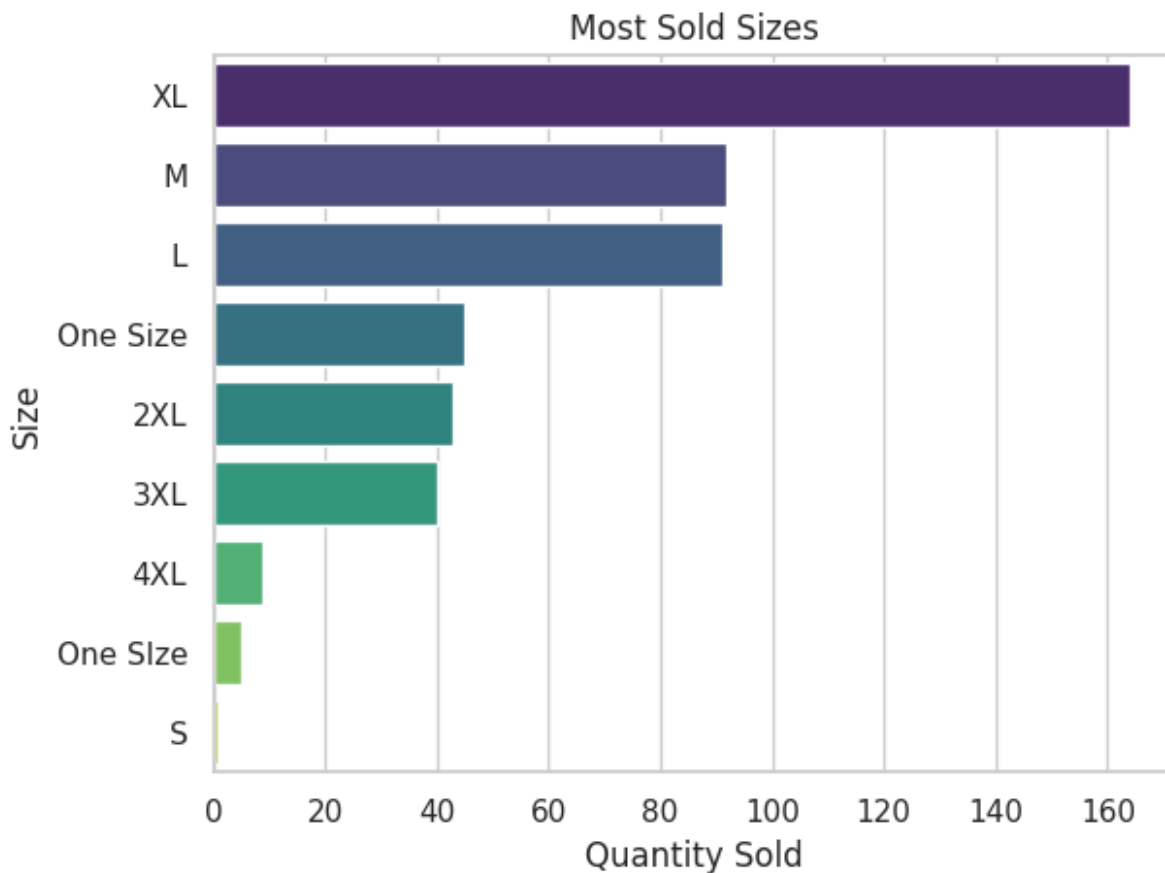


```
sns.set(style="whitegrid")
sns.barplot(x=size_quantity.values, y=size_quantity.index,
palette="viridis")
plt.title("Most Sold Sizes")
plt.xlabel("Quantity Sold")
plt.ylabel("Size")
plt.show()
```

<ipython-input-40-d592c37c79f2>:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x=size_quantity.values, y=size_quantity.index,
palette="viridis")
```

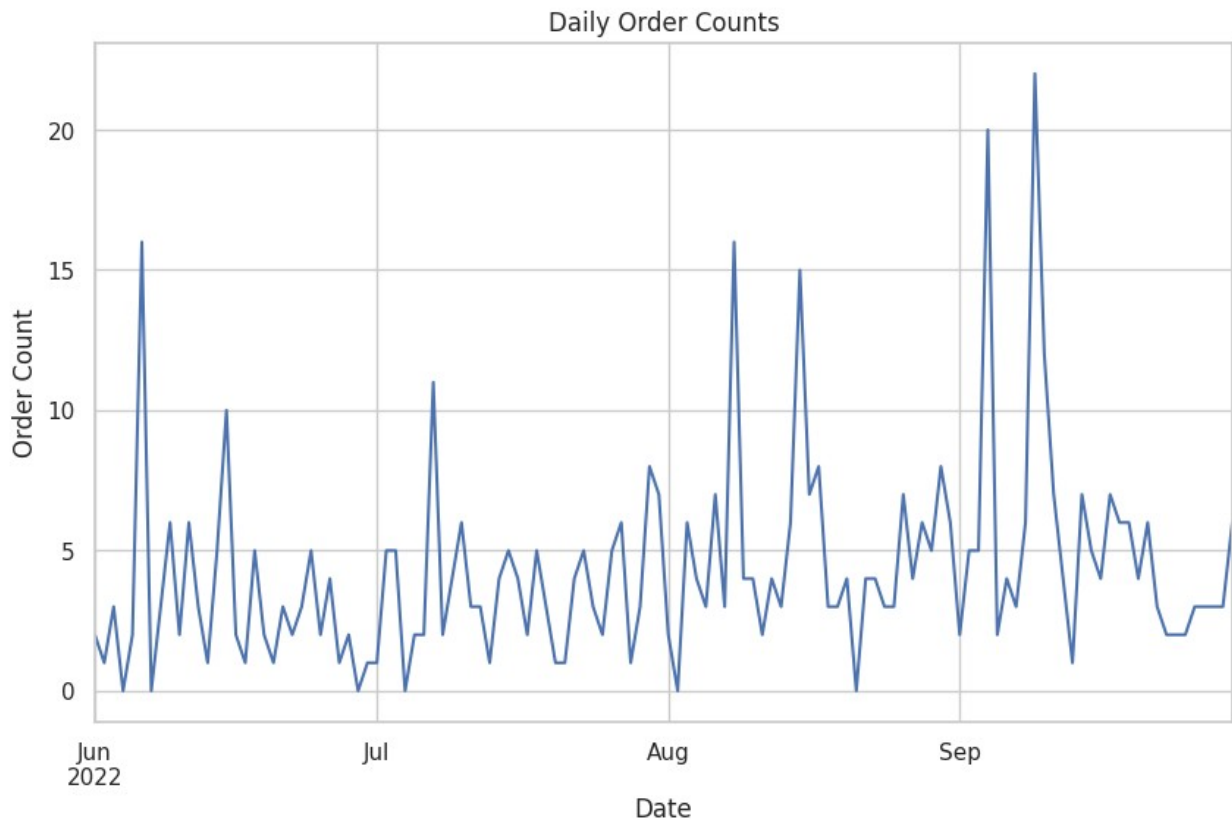


Analyze daily order counts

```
daily_orders = dataset.resample('D', on='order_date')  
['order_id'].count()  
print(daily_orders)
```

```
order_date  
2022-06-01    2  
2022-06-02    1  
2022-06-03    3  
2022-06-04    0  
2022-06-05    2  
...  
2022-09-26    3  
2022-09-27    3  
2022-09-28    3  
2022-09-29    3  
2022-09-30    6  
Freq: D, Name: order_id, Length: 122, dtype: int64  
  
daily_orders.plot(kind='line', figsize=(10, 6))  
plt.xlabel('Date')  
plt.ylabel('Order Count')
```

```
plt.title('Daily Order Counts')
plt.show()
```



find duplicate rows

```
duplicates = dataset [dataset.duplicated()]
print(f"Number of duplicate rows: {len(duplicates)}")
duplicates.head()
```

Number of duplicate rows: 0

```
{"repr_error": "Out of range float values are not JSON compliant:
nan", "type": "dataframe", "variable_name": "duplicates"}
```

Detect high-revenue outliers

```
Q1 = dataset['revenue'].quantile(0.25)
Q3 = dataset['revenue'].quantile(0.75)
IQR = Q3 - Q1
```

```
# Define upper bound for outliers
```

```
upper_bound = Q3 + 1.5 * IQR
```

```
# Find rows where revenue exceeds the upper bound
```

```

high_revenue_outliers = dataset[dataset['revenue'] > upper_bound]
print(f"Number of high-revenue outliers:
{len(high_revenue_outliers)}")
high_revenue_outliers.sort_values(by='revenue',
ascending=False).head()

```

Number of high-revenue outliers: 7

```

{"summary":{"\n  \"name\": \"high_revenue_outliers\",\n  \"rows\": 5,\n  \"fields\": {\n    \"column\": \"order_id\",\n    \"properties\": {\n      \"dtype\": \"number\",\n      \"std\": 87,\n      \"min\": 15,\n      \"max\": 242,\n      \"num_unique_values\": 5,\n      \"samples\": [\n        164,\n        83,\n        84\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"order_date\",\n    \"properties\": {\n      \"dtype\": \"date\",\n      \"min\": \"2022-07-06 17:14:00\",\n      \"max\": \"2022-09-19 10:05:00\",\n      \"num_unique_values\": 5,\n      \"samples\": [\n        \"2022-07-06 17:14:00\",\n        \"2022-09-19 10:05:00\",\n        \"2022-07-11 17:19:00\"\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"sku\",\n    \"properties\": {\n      \"dtype\": \"string\",\n      \"num_unique_values\": 3,\n      \"samples\": [\n        \"708\",\n        \"9699\",\n        \"799\"\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"color\",\n    \"properties\": {\n      \"dtype\": \"category\",\n      \"num_unique_values\": 2,\n      \"samples\": [\n        \"Black\",\n        \"Dark Blue\"\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"size\",\n    \"properties\": {\n      \"dtype\": \"string\",\n      \"num_unique_values\": 5,\n      \"samples\": [\n        \"One Size\",\n        \"L\"\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"unit_price\",\n    \"properties\": {\n      \"dtype\": \"number\",\n      \"std\": 273,\n      \"min\": 288,\n      \"max\": 899,\n      \"num_unique_values\": 2,\n      \"samples\": [\n        288,\n        899\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"quantity\",\n    \"properties\": {\n      \"dtype\": \"number\",\n      \"std\": 0,\n      \"min\": 1,\n      \"max\": 2,\n      \"num_unique_values\": 2,\n      \"samples\": [\n        2,\n        1\n      ],\n      \"semantic_type\": \"\",\n      \"description\": \"\"\n    },\n    \"column\": \"revenue\",\n    \"properties\": {\n      \"dtype\": \"number\",\n      \"std\": 144,\n      \"min\": 576,\n      \"max\": 899,\n      \"num_unique_values\": 2,\n      \"samples\": [\n        576,\n        899\n      ],\n    }

```

```
\nsemantic_type\": \"\", \n\n\"description\": \"\" \n\n}\n}\n]\", \"type\": \"dataframe\"}
```

visulize with a box plot

```
sns.boxplot(data=dataset, x='revenue', color='lightcoral')  
plt.title("Revenue Distribution with Outliers")  
plt.show()
```

